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Revealing the atomistic nature of dislocation-precipitate interactions in Al-Cu alloys

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ABSTRACT

Despite significant gains on understanding strengthening mechanisms in precipitate strengthened materials, such as aluminum alloys, there persists a sizeable gap in the atomistic understanding of how different precipitate types and their morphology along with dislocation character affects the hardening mechanisms. Toward this, the paper examines nature of precipitation strengthening behavior observed in the Al-Cu alloys using atomistic simulations. Specifically, the critical resolved shear stress is quantified across a wide range of dislocation-precipitate interactions scenarios for both θ' and θ phase of Al_2Cu. Overall, the simulations reveal that the dislocation character (edge or screw) plays a key role in determining the predominant hardening mechanism (shearing vs. Orowan looping) employed to overcome the θ' Al_2Cu precipitate. Furthermore, the critical shear stress and mechanism to overcome the precipitate is sensitivity to the position of the glide plane with respect to the precipitate and its orientation. Interestingly in our findings, the θ Al_2Cu precipitate conventionally regarded as un-shearable particle was overcome by shear cutting mechanism for small equivalent precipitate radius, which agrees with recent TEM observations. These findings provide necessary information for the development of atomistically informed precipitate hardening models for the traditional continuum scale modeling efforts.

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1. Introduction

Precipitation-strengthened alloys, such as aluminum alloys, are ubiquitously utilized for a wide range of structural applications due to their superior mechanical performance derived from the complex distribution of precipitates across the microstructure. These precipitates serve as effective obstacles to dislocation motion, thereby imparting increase in flow strength. The mechanical behavior of these precipitate-strengthened alloys is controlled by the nature of the precipitate/matrix interface, precipitate morphology (size, shape, aspect ratio, etc.), and subsequent dislocation-precipitate interactions [1]. If the precipitates are significantly smaller than the dislocation line length and the interface is coherent, dislocations typically overcome the precipitate by shearing. The critical resolved shear stress (τ_c) required for precipitate shearing to occur can be expressed in the following general form [2–4]:

$$\tau_c = \frac{2}{b\sqrt{\Gamma}}\left(\frac{1}{\lambda}\right)\left(\frac{F}{2}\right)^{3/2} \tag{1a}$$

$$F = kr_{eq}^m \tag{1b}$$

$$\Gamma = \frac{Gb^2}{4\pi}\left[\frac{1+\nu-3\sin^2\beta}{1-\nu}\right]\ln\left(\frac{\chi}{r_0}\right) \tag{1c}$$

where, b is the Burger's vector magnitude of the dislocation line, λ is the mean planar spacing between precipitates, F is the resistive force experienced by the dislocation, r_{eq} is the equivalent precipitate radius on the slip plane, k is a parameter that is decided based on the nature of strengthening mechanism [2,3], m is a constant that varies between 0 and 1 for the different strengthening mechanism observed during precipitate shearing, β is the angle between the Burger's vector and dislocation line (i.e., for an edge dislocation, $\beta = \pi/2$, and for a screw dislocation, $\beta = 0$), χ and r_0 are the outer and inner cut-off radii around the dislocation core required to approximate the line energy (Γ), and ν is the Poisson's ratio.

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On the other hand, in the overaged condition, the precipitates are generally larger in size and incoherent with the matrix, therefore the dislocation overcomes the precipitate by looping. Several analytical solutions based on the original Orowan mechanism [5] exist that correlate the shear stress required to bypass the dislocation with the spacing between particles (λ), but the generalized formulation can be expressed in the following manner:

$$\tau_c = \xi \frac{Gb}{2\pi\lambda} \ln\left(\frac{\chi}{r_0}\right) \tag{2}$$

where, ξ is 1 for the edge dislocation and $(1/(1-\nu))$ for the screw dislocation.

Despite these efforts to develop a relationship to quantify critical resolved shear stress for various precipitate strengthening mechanisms, the complex effects of the precipitate shape, precipitate orientation, variation in interaction force due to the different positions between the dislocation and precipitate, and dislocation character were unaccounted for in these models. Subsequently, this motivated several theoretical studies [6–10] to account for these complex effects; however, such efforts have still been unable to establish a general consensus with experimental findings on these complex interactions during precipitation hardening [6,11–13].

Atomistic simulations can serve as a viable tool in understanding the influence of various geometric and mechanistic variables on the precipitate strengthening mechanism. In the past, atomistic tools have been utilized to study the strengthening due to coherent precipitates [14–16], dislocation interaction with different solutes, void and precipitate in alpha-iron [17–19], and interaction of dislocation with Guinier-Preston (GP) zones in Al-Cu alloys [20–22]. Furthermore, the quantification of the interfacial energies between Al and θ' Al_2Cu and its implications on the precipitate growth behavior were examined using density functional theory (DFT) calculations and phase field modeling [23,24]. Similarly, discrete dislocation dynamics has been employed to reveal the effect of solute friction on the dislocation-dislocation interactions [25], the role of dislocation character on bypass and pinning mechanisms [26], and the effect of source and obstacle strengths on yield stress [27]. Despite these efforts, there is a lack of clarity on the strengthening mechanisms in these alloys and the influence of various microstructural features (precipitate spacing, size, orientation, interfacial energy, etc.) on the macroscopic flow stress.

In this work, we systemically examine the nature of precipitation strengthening behavior observed in the Al-Cu alloys using atomistic simulation. Specifically, the critical resolved shear stress (τ_c) was quantified across a wide range of dislocation-precipitate interactions scenarios for both θ' and θ phase of Al_2Cu. The paper is organized in the following manner: the next section describes the computational methods for exploring the dislocation-precipitate interactions. Then, we present the atomic insights of various factors on the hardening mechanism achieved by θ' and θ precipitates of Al_2Cu. For instance, we show that the dislocation character (edge or screw) plays a key role in determining the predominant hardening mechanism (shearing vs. Orowan looping) employed to overcome the θ' Al_2Cu precipitate. Furthermore, the critical shear stress and mechanism to overcome the precipitate was influenced by the position of the glide plane with respect to the precipitate and its orientation. These findings provide necessary information for the development of atomistically informed precipitate hardening models [28] and mesoscale modeling efforts [29] towards predicting more accurate mechanical response and behavior of precipitate strengthened alloys.

2. Computational methods

The atomistic simulations were carried out using the angular dependent interatomic potential developed by Apostol and Mishin [30] within the LAMMPS framework [31]. The interatomic potential was able to accurately predict the structural properties (lattice parameter, elastic constant, formation energies and surface energies) for the θ' and θ phases of Al_2Cu (for details refer [30]). The simulation cell dimensions were approximately 250 Å × 250 Å × 750 Å, see Fig. 1. In the case of edge dislocation, the Al atoms (face centered cubic lattice) were oriented along the X $[\bar{1}01]$, Y [111] and Z $[\bar{1}2\bar{1}]$ directions (Fig. 1a). Similarly, for the screw dislocation the Al atoms were oriented along the X $[11\bar{2}]$, Y [111] and Z $[1\bar{1}0]$ directions (Fig. 1b). The dislocations were introduced by applying the Stroh's anisotropic displacement field [32]. The θ' phase of the Al_2Cu precipitate typically has a rod like shape with the following orientation relationship $(001)_{\theta'} \,||\, \{001\}_{Al}$ (i.e., the precipitate undergoes larger growth along the c axis) [23]. The Al_2Cu θ can precipitate in several different orientations (>100) with respect to the Al matrix [33]. In this work, we have examined the Guinier I orientation relationship, i.e., $(001)_\theta \,||\, (001)_{Al}$, $[100]_\theta \,||\, [110]_{Al}$. Therefore, there are two unique interaction scenarios between the dislocation and the θ' *and* θ precipitates based on the angle between the Burger's vector and precipitate habit plane. For instance, the Burger's vector along the $[\bar{1}01]$ direction lying on the (010) habit plane yields a 0° offset. On the other hand, the Burger's vector along the $[1\bar{1}0]$ direction lying on the (010) habit plane results in a 60° offset. The θ' phase of Al_2Cu precipitate with approximately 16 Å × 16 Å × 28 Å dimensions along the

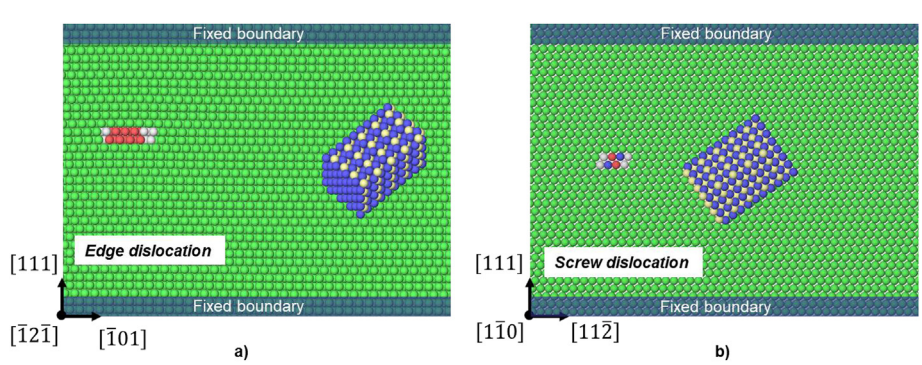

Fig. 1. The simulation cell used to study the a) edge and b) screw dislocations interaction with precipitate in Al. The Al atoms in the matrix are colored in green with red atoms representing the stacking fault. The Al_2Cu precipitate atoms were colored blue (Al) and golden (Cu). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

$\{100\}_{Al}$ direction was introduced within the matrix to form the interfaces described in Fig. 1. Similarly, the θ phase of Al_2Cu precipitate with approximately 24 Å × 24 Å × 25 Å dimensions was created based on the Guinier type I orientation relationship described earlier. The precipitate dimensions were chosen to minimize the lattice mismatch strain while keeping the aspect ratio for θ' Al2Cu precipitate close to the experimental observations [34]. The free boundary condition was prescribed along the Y direction; whereas, a periodic boundary condition was maintained along the X and Z directions (Fig. 1). Further, the atomistic model was then equilibrated at a temperature of 300 K using a micro-canonical ensemble (NVE) for 20 ns, and the periodic direction along the Z direction was subsequently relaxed using the isothermal-isobaric (NPT) equations of motion for 30 ns. The top and bottom boundary atoms along the Y direction were fixed (Fig. 1) and an incremental shear displacement was applied to the top of these atoms in the X and Z directions for the edge and screw dislocations, respectively, to obtain a constant shear strain rate of 10^8 s^{-1}. The results were visualized with the help of OVITO [35].

3. Results

3.1. Interaction of the edge dislocation with the θ' Al_2Cu precipitate

The interaction between the edge dislocation and θ' Al_2Cu precipitate was first examined with the help of common neighbor analysis (CNA) [36] and atomic von Mises strain [37,38] (Fig. 2a–e). The thermal activation energy available at 300 K and the elastic interactions due to the presence of a precipitate were found to be insufficient to initiate dislocation glide (~3 MPa) for the various cases examined here without the external load. However, the application of incremental shear strain (ε_{xy}) was found to cause the dislocation to glide towards the precipitate and begin to bow around the precipitate (Fig. 2a and b). Firstly, the 0° offset orientation with an equivalent precipitate radius of 12.5 Å was examined

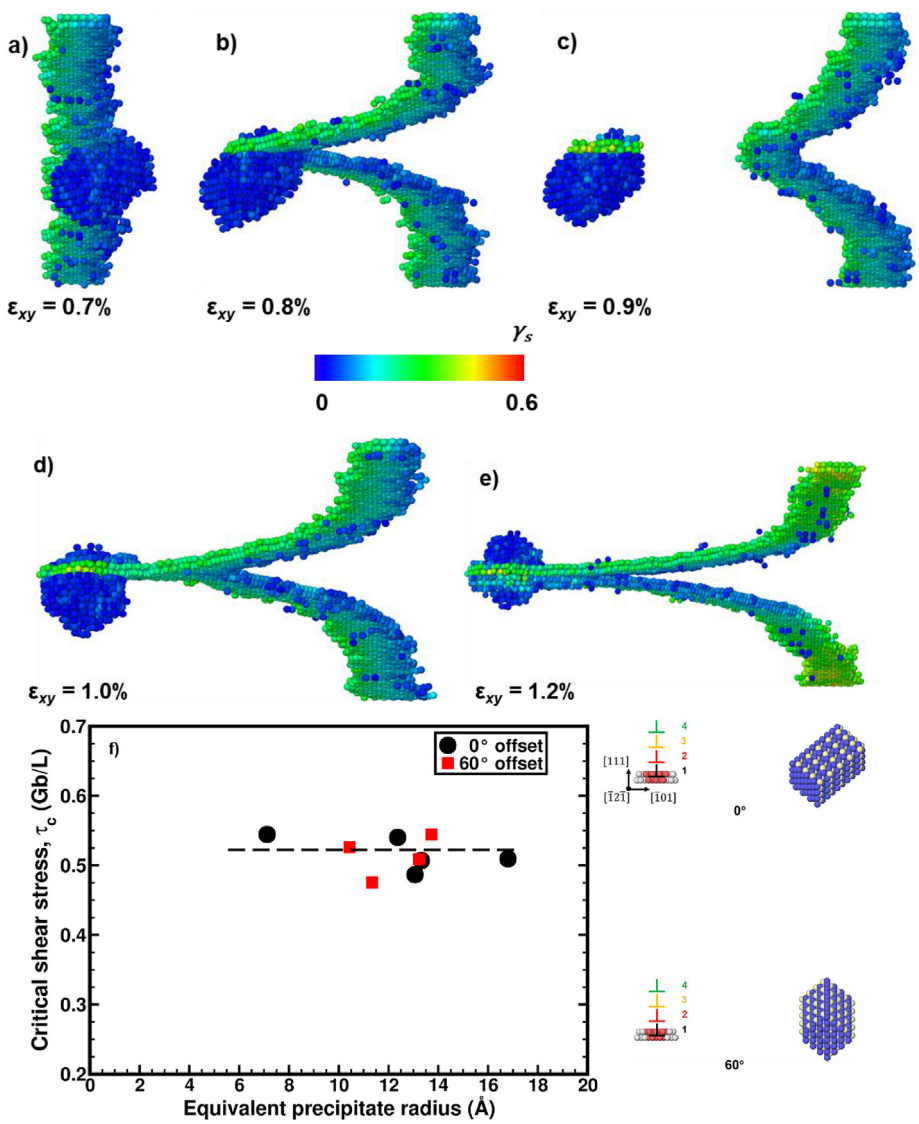

Fig. 2. The interaction between an edge dislocation and θ' Al_2Cu precipitate for: a-c) the 0° offset precipitate with an equivalent precipitate radius of 12.5 Å scenario; and d-e) the 60° offset precipitate with an equivalent precipitate radius of 13.8 Å scenario. The atoms are colored according to the von Mises shear strain and only the dislocation line and precipitate atoms are shown (based on the CNA). e) The influence of the precipitate orientation and glide plane on the critical shear stress (τ_c) required for the edge dislocation to overcome the θ' Al_2Cu precipitate. The shear stress was normalized by the Orowan's shear stress $\left(\tau_{or} = \frac{Gb}{\lambda} = 120\ MPa\right)$, where G is the shear modulus for Al (31.6 GPa), b is the Burger's vector magnitude (2.86 Å) and λ is the precipitate spacing (~750 Å). Note that the 1 to 4 represent the vertical offset between the center of precipitate and the initial edge dislocation slip plane.

by varying the dislocation plane with respect to the θ' Al_2Cu precipitate. In this case, at a shear stress of 65 MPa or $0.54\tau_{or}$ (where $\tau_{or}=Gb/\lambda = 120.6$ MPa) the dislocation was found to overcome the precipitate by looping around it (Fig. 2c). Similarly, the edge dislocation for the 60° offset orientation and an equivalent precipitate radius of 13.8 Å overcame the precipitate by bowing around it at a shear stress of 65 MPa or $0.54\tau_{or}$ (Fig. 2d and e). In a similar manner, the effect of the θ' Al_2Cu precipitate orientation (0° and 60° offset) and the equivalent radius on the shear stress required for the dislocation to overcome the precipitate was examined across various scenarios (Fig. 2f). Interestingly, both the orientation and equivalent precipitate radius had negligible effect on the critical shear stress (τ_c) required by the dislocation to overcome the precipitate. Furthermore, the edge dislocation overcame the θ' Al_2Cu precipitate predominantly by the Orowan looping and only for the smaller equivalent precipitate radius (6.5 Å) shear cutting was observed.

3.2. Interaction of the screw dislocation with the θ' Al_2Cu precipitate

In general, most precipitate hardening models examine the dislocation character by accounting the line tension (Equation (1c)) or the parameter ξ for non-shearable θ' Al_2Cu precipitates (Equation (2)). Therefore, to understand the atomistic mechanisms involved by the screw dislocation to overcome the θ' Al_2Cu precipitate different scenarios (precipitate orientation and equivalent radius) were examined here (Fig. 3). For instance, in the case of the

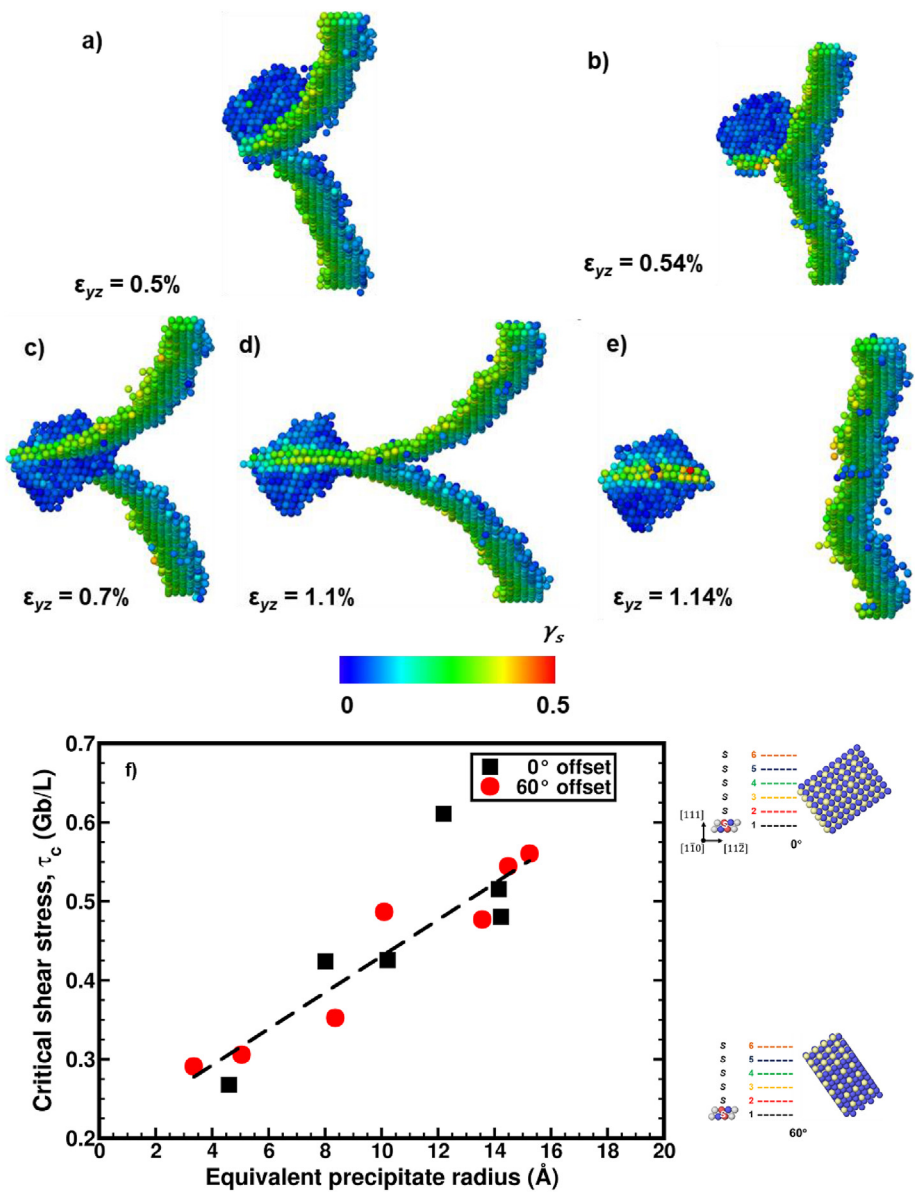

Fig. 3. The interaction between a screw dislocation and θ' Al_2Cu precipitate for: a-c) the 0° offset precipitate with an equivalent precipitate radius of 4.6 Å scenario; and d-e) the 60° offset precipitate with an equivalent precipitate radius of 14.1 Å scenario. The atoms are colored according to the von Mises shear strain and only the dislocation line and precipitate atoms are shown (based on the CNA). The influence of the precipitate orientation and glide plane on the critical shear stress (τ_c) required for the screw dislocation to overcome the θ' Al_2Cu precipitate. The shear stress was normalized by the Orowan's shear stress ($\tau_{or} = 120\ MPa$). Note 1 to 6 represent the vertical offset between the center of precipitate and the initial screw dislocation slip plane.

0° offset with an equivalent precipitate radius of 4.6 Å, the screw dislocation overcame at a $\tau_c = 32$ MPa ($0.26\tau_{or}$) by shear cutting (Fig. 3a and b). However, for the 60° orientation and an equivalent precipitate radius of 14.1 Å, the dislocation was found to loop around the precipitate (Fig. 3c–e). Interestingly, the screw dislocation predominantly overcame the θ' Al_2Cu precipitate by shearing cutting regardless of the precipitate orientation, except for scenarios when the equivalent precipitate radius was larger than 14.5 Å. Furthermore, the critical shear stress (τ_c) required to overcome the precipitate was found to vary linearly with the equivalent precipitate radius (Fig. 3f). In terms of the hardening model, assuming a value of 2/3 for the parameter m would yield in a linear relationship between the critical shear stress (τ_c) and equivalent precipitate radius (r_{eq}) (Equation (1)).

3.3. Interaction of the edge dislocation with the θ Al_2Cu precipitate

The presence of non-deformable precipitates provides the maximum strengthening behavior in precipitate strengthened materials. Experimentally, it has been observed that the θ Al_2Cu precipitate forms an incoherent interface with the Al matrix, thereby forcing the dislocation to loop around [39,40]. Here, the

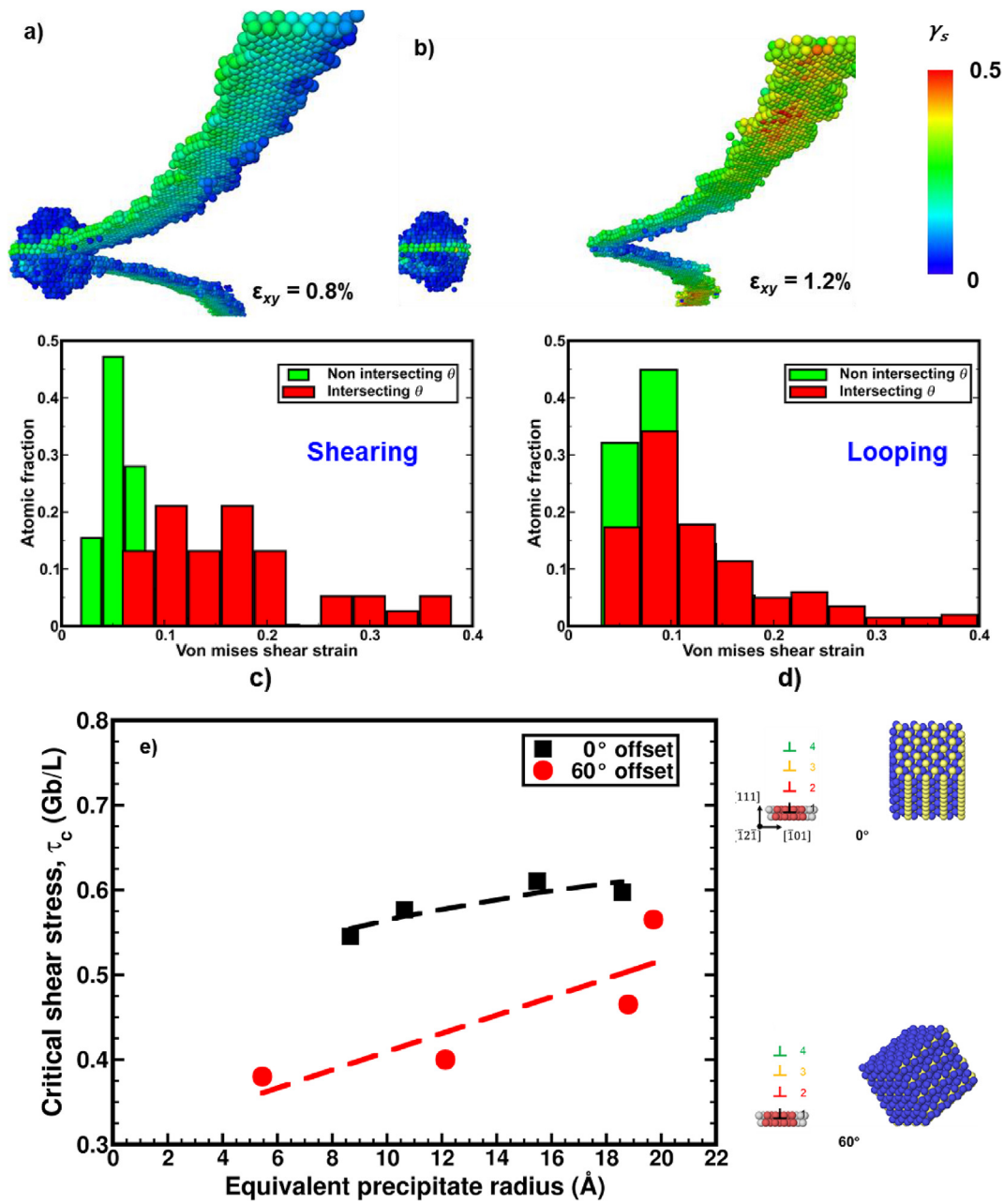

Fig. 4. a-b) The interaction between an edge dislocation and θ Al_2Cu precipitate for the 60° offset precipitate with an equivalent precipitate radius of 10 Å scenario. The atoms are colored according to the von Mises shear strain and only the dislocation line and precipitate atoms are shown (based on CNA). The variation in von Mises equivalent strain for intersecting and non-intersecting atoms within the Al_2Cu in the case of: c) shearing; and d) Orowan looping mechanisms. e) The influence of the precipitate orientation and glide plane on the critical shear stress (τ_c) required for the edge dislocation to overcome the θ Al_2Cu precipitate. The shear stress was normalized by the Orowan's shear stress (τ_{or} = 120 *MPa*).

atomic scale observations also found that the edge dislocation overcame the θ Al_2Cu precipitate predominantly (regardless of the precipitate orientation) by the Orowan looping mechanism (Fig. 4a–b). However, despite the incoherent interfaces shear cutting was observed for scenarios when the equivalent precipitate radius was smaller than 10 Å. The von Mises shear strain distribution within the θ Al_2Cu precipitate for the atoms that intersected with the slip plane was compared against the non-intersecting atoms which are above and below the slip plane to clearly identify the shear cutting mechanism (Fig. 4c–d). The critical shear stress (τ_c) increased with larger intersecting precipitate area (~6 MPa for 0° offset, ~20 MPa for 60° offset, Fig. 4e). Furthermore, the critical shear stress (τ_c) response was dependent on the precipitate orientation (Fig. 4e). This can be attributed to the evolving interfacial energy obtained by varying the dislocation position for the 60° offset orientation. However, for the 0° offset the dislocation line encounters only the higher energy interface (refer A2). Therefore, interfacial strengthening is a plausible mechanism to

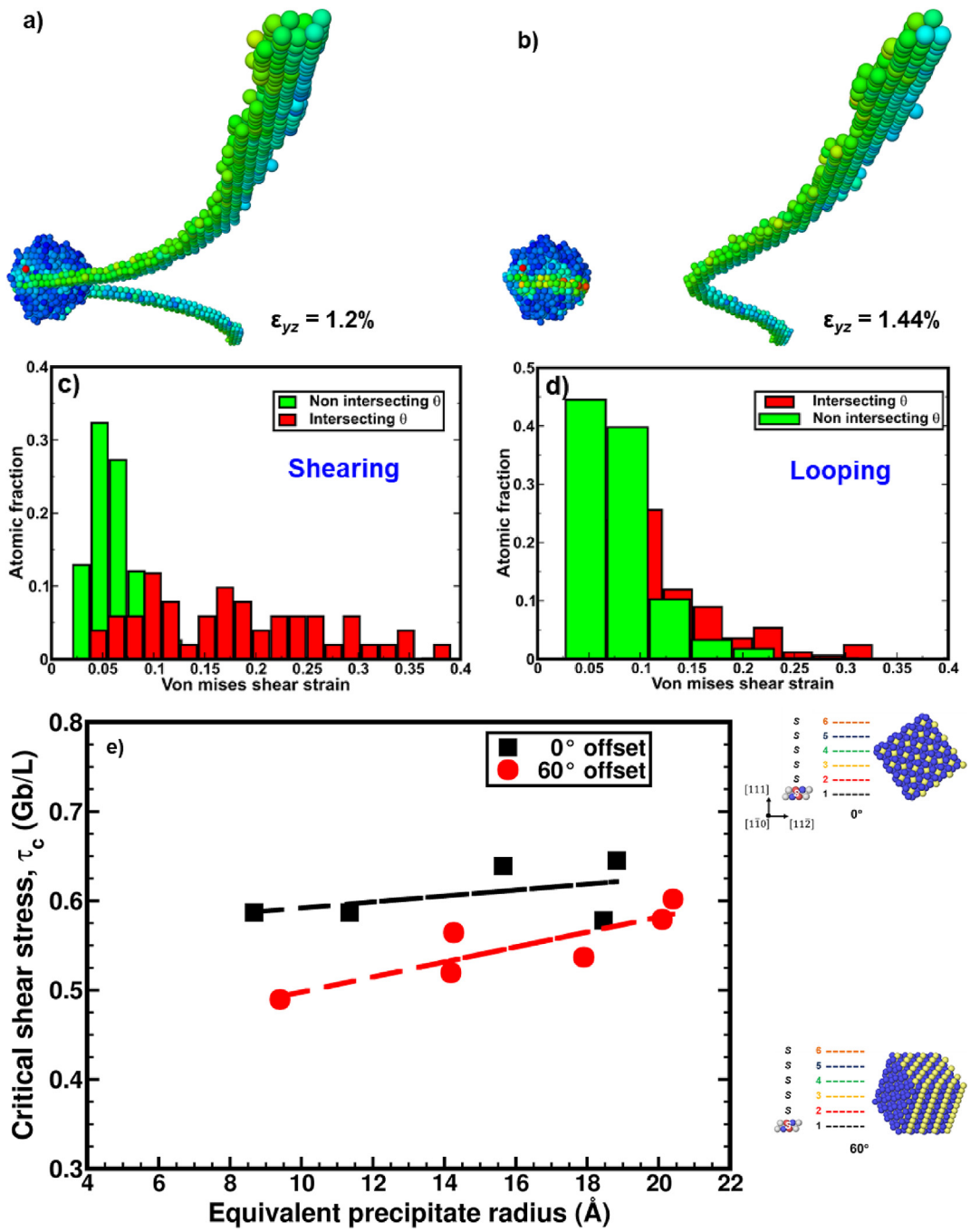

Fig. 5. a-b) The interaction between a screw dislocation and θ Al_2Cu precipitate for the 0° offset precipitate with an equivalent precipitate radius of 18.8 Å scenario. The atoms are colored according to the von Mises shear strain and only the dislocation line and precipitate atoms are shown (based on CNA). The variation in von Mises equivalent strain for intersecting and non-intersecting atoms within the Al_2Cu in the case of: c) shearing; and d) Orowan looping mechanisms. e) The influence of the precipitate orientation and glide plane on the critical shear stress (τ_c) required for the screw dislocation to overcome the θ Al_2Cu precipitate. The shear stress was normalized by the Orowan's shear stress (τ_{or} = 120 *MPa*).

explain the observed critical shear stress response between the edge dislocation and θ Al_2Cu precipitate.

3.4. Interaction of the screw dislocation with the θ Al_2Cu precipitate

Based on the line tension expression (Equation (1c)), we understand that the screw dislocation possess almost four times larger line tension when compared to an edge dislocation (assuming, $\nu = 1/3$). In terms of precipitate hardening behavior, this would imply that the screw dislocation line would be less flexible during an Orowan looping as compared to an edge dislocation. Therefore, the influence of the precipitate orientation and screw dislocation glide plane on critical the shear stress (τ_c) required to overcome the θ Al_2Cu precipitate was assessed across different scenarios. The atomic insights revealed that the screw dislocation overcame the precipitate predominantly by the Orowan looping (Fig. 5a–b). However, the shear cutting was observed for scenarios where the equivalent precipitate radius was less than 10 Å. This was verified by comparing the von Mises shear strain for the intersecting and non-intersecting Al_2Cu atoms (Fig. 5c–d). In the case of 0° offset, the critical shear stress (τ_c) was independent of the equivalent precipitate radius (Fig. 5e). On the other hand, in the case of 60° offset demonstrated a slight increase in the critical shear stress (τ_c) with equivalent precipitate radius (Fig. 5e). This can be attributed to the varying strain field at the matrix-precipitate interface encountered by the dislocation line for the 60° offset precipitate orientation.

4. Discussion

The precipitate strengthening mechanisms for Al-Cu alloys consisting of θ' and θ Al_2Cu inter-metallics were studied employing atomistic simulations. In other words, a wide range of dislocation precipitate interactions scenarios for both θ' and θ phase of Al_2Cu were examined systematically to understand the influence of precipitate orientation, position of the glide plane with respect to the precipitate and dislocation character. Firstly, our findings revealed that it was quite complicated to find a τ_c that represents an ensemble average of all the various possible interactions as there is a large degree of interplay between the variables. Nonetheless, in the case of θ' the predicted averages for τ_c (approximately 60 MPa) were found to be in good agreement with experimental findings for under aged and overaged samples [7]. Furthermore, the precipitate hardening mechanism for θ' was found to depend on the dislocation character. However, microstructural analysis of deformed Al-Cu samples revealed that θ' particles were overcome by Orowan looping mechanism [41]. During the aging process of Al-Cu alloys, θ particles nucleate on existing θ' plates or in the vicinity of θ' plates. These θ nuclei then grow by causing dissolution of the nearby θ' plates. Further, during the aging, θ particles continue to grow resulting in an increased inter-particle spacing. At this stage, the particles do not serve as an effective barrier to dislocation motion because the critical shear stress to bow around the obstacle decreases (refer Equation (2)). Therefore, the θ particles do not contribute to the initial yield behavior, but the hardening behavior of overaged Al-Cu alloys improves as θ particles store Orowan loops around them. This leads to an increase in secondary work hardening of precipitate strengthened Al-Cu alloys. In our work, we maintained the same interparticle spacing for examining dislocation precipitate interactions for θ' and θ particles. Additionally, the θ and θ' particles were of similar size. This recreated a scenario of smaller θ particles with semi-coherent interfaces, thereby causing variation in critical resolved shear stress, as the precipitate orientation was altered (Figs. 4 and 5).

5. Conclusions

The following conclusions can be drawn on the several different dislocation-precipitate interactions scenarios (obtained by varying precipitate orientation and dislocation glide plane) examined here:

1. In the case of the edge dislocation-θ' interaction, the critical shear stress was unaffected by the precipitate orientation and area. Furthermore, the dislocation predominantly employed Orowan looping mechanism to overcome the precipitate.
2. In the case of the screw dislocation-θ' interaction, the screw dislocation sheared through the θ' precipitate in majority of the scenarios assessed by coherent strengthening mechanism. The overall critical shear stress response was found to vary linearly with increasing equivalent precipitate radius. This behavior agrees with the classical hardening model for shearable precipitates. Additionally, these results clearly highlight the influence of the dislocation character.
3. In the case of dislocation-θ interaction, the dislocation (both edge and screw) predominantly overcame the precipitate by the Orowan looping mechanism. However, contrary to consensus shear cutting mechanism was also observed in scenarios when the equivalent radius was small. Furthermore, the precipitate orientation was found to have a distinct effect on the critical shear stress (regardless of the dislocation character).

Overall, in this work dislocation-precipitate interactions were examined across several scenarios obtained by varying the dislocation character, precipitate orientation (angle between the Burgers vector and precipitate habit plane) and dislocation glide plane w.r.t the precipitate. Interestingly, in the case of the θ' Al_2Cu precipitate it was found that a change in the dislocation character results in a shift of the dominant mechanism ($\perp \rightarrow S$, looping $\rightarrow$ cutting). Furthermore, it was found that the θ Al_2Cu precipitate conventionally regarded as un-shearable particle can be overcome by shear cutting mechanism in agreement with recent TEM observations [42]. In summary, these findings along with the work of Singh et al. [20,21] provides the necessary atomistic insights for the development of an age hardening model.

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Appendix

A.1 Modeling the θ' phase of the Al_2Cu precipitate

The θ' phase of Al_2Cu has a tetragonally distorted cubic fluorite structure ($Fm\overline{3}m$, CaF_2) (Fig. A1a) with the following orientation relationship with the Al matrix $(001)_{\theta'} \left|\right| \{001\}_{Al}$. The θ' precipitate forms a coherent interface with the Al matrix along the $(001)_{\theta'} \left|\right| (001)_{Al}$ and semi-coherent interfaces along the $(100)_{\theta'} \left|\right| (100)_{Al}$ and $(010)_{\theta'} \left|\right| (010)_{Al}$ (Fig. A1b-c). The interface energy for the coherent and the semi-coherent interfaces were found to be 265 mJ/m^2 and 702 mJ/m^2, respectively were found to be in reasonable agreement with previous findings [23,24].

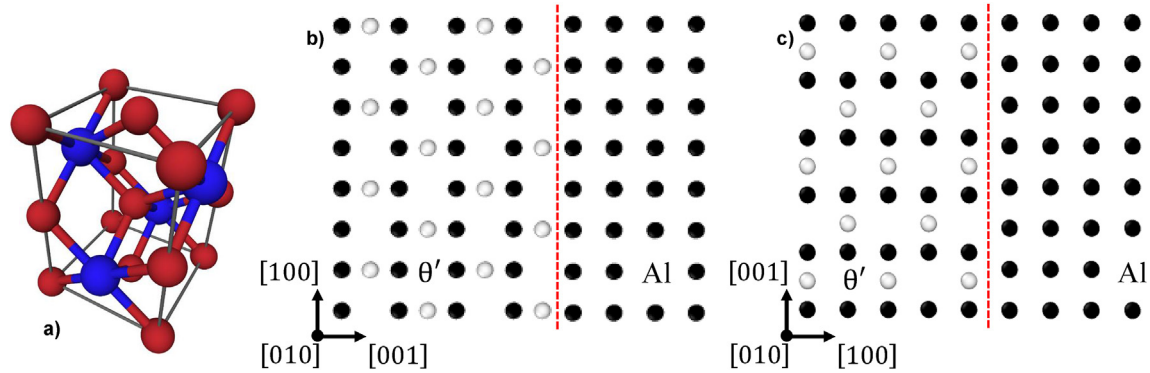

Fig. A1. a) The tetragonally distorted cubic fluorite structure of the θ' phase of an Al_2Cu precipitate. The atoms in red and blue represent the position of Al and Cu atoms, respectively. The b) coherent and c) the semi-coherent interfaces formed between the θ' Al_2Cu precipitate and the Al matrix. The black and the white dots represent the Al and Cu atoms, respectively.

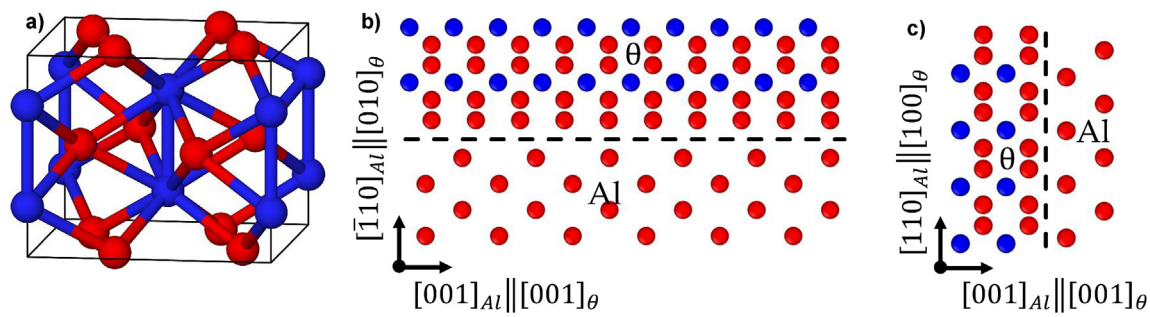

Fig. A2. a) The unit cell for the θ phase of an Al_2Cu precipitate. The atoms in red and blue represent the position of Al and Cu atoms, respectively. The semi-coherent interfaces b) $(\bar{1}10)_{Al}|(010)_{\theta}$ and c) $(001)_{Al}|(001)_{\theta}$ formed between the θ Al_2Cu precipitate and the Al matrix.

A.2 Modeling the θ phase of the Al_2Cu precipitate

The interactions of edge and screw dislocations with an Al_2Cu θ precipitate for different crystallographic orientations were examined using atomistic simulations [31]. The θ Al_2Cu has a body centered tetragonal structure ($I4/mcm$) as shown in Fig. A2a. Furthermore, due to the high degree of misfit between the Al_2Cu θ and the Al matrix the precipitate can nucleate at several different orientations (>100) with respect to the Al matrix [33]. In this work, we have considered the Guinier I relationship:

$$(001)_{\theta}||(001)_{Al},\ [100]_{\theta}||\ [110]_{Al}$$

Thus, both the interfaces formed between θ and Al matrix are semi-coherent (Fig. A2b-c). The interface energies for both the semi-coherent interfaces were found to be 663.66 mJ/m^2 (Fig. A2b) and 568.25 mJ/m^2 (Fig. A2c), respectively.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jallcom.2019.05.110.

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